

Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

TEAM ID	SWTID-2026-3007
PROJECT TITLE	Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau
MAX MARKS	5 MARKS

4.3 – Solution Architecture

The Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau project follows a structured solution architecture that transforms raw cosmetic product data into meaningful business insights through data visualization and analytics. The architecture consists of multiple layers, including data collection, data preparation, data analysis, visualization, and reporting. These layers work together to provide a comprehensive understanding of cosmetic market trends, consumer preferences, and brand performance.

The first layer is the Data Source Layer, where cosmetic product data is collected from Excel spreadsheets or CSV files. The dataset contains important information such as product names, brands, categories, prices, ratings, reviews, ingredients, and skin-type suitability. This data serves as the foundation for all analytical activities within the project.

The second layer is the Data Preparation Layer, where the collected data is cleaned, validated, and organized. Missing values, duplicate records, and inconsistencies are removed to improve data quality and ensure accurate analysis. Proper data preparation helps create reliable visualizations and meaningful insights.

The third layer is the Analytics Layer, which uses Tableau Desktop to process and analyse the prepared data. Various analytical techniques are applied to evaluate category-wise product distribution, top-performing brands, average product prices, skin-type suitability, customer review patterns, and market trends. Tableau's calculated fields, filters, and parameters enhance analytical capabilities and allow deeper exploration of the data.

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Overall, the solution architecture provides a systematic framework for collecting, processing, analysing, and visualizing cosmetic industry data. By leveraging Tableau's powerful business intelligence features, the architecture transforms raw data into actionable insights, enabling organizations to make data-driven decisions and gain a competitive advantage in the cosmetics industry.

